SYED AZMUL HASAN

Data Analyst | Business Analyst | AI/ML Researcher | Finance & Operations Enthusiast

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LinkedIn • GitHub • **Portfolio**

PROFESSIONAL SUMMARY

Mathematics graduate with expertise in AI/ML, data analytics, and statistical analysis, complemented by research experience in deep learning and computer vision. Developed high-performing AI models, contributed to research publications, and demonstrated leadership, communication, and problem-solving skills through academic and organizational activities. Seeking to apply analytical and data-driven decision-making abilities in technology and corporate environments.

CORE COMPETENCIES

- Artificial Intelligence
- Computer Vision
- Statistical Analysis
- Technical Writing
- Business Analysis
- Machine Learning
- Data Analytics
- Project Management
- Leadership
- Data-Driven Decision Making
- Deep Learning
- Research & Development
- Problem Solving
- Team Collaboration
- Microsoft Office Suite

TECHNICAL SKILLS

Programming & Databases: Python (Intermediate) • MATLAB (Intermediate) • C/C++ (Intermediate) • SQL (Intermediate) • Java (Beginner) • R (Beginner)

Artificial Intelligence & Data Science: Machine Learning (Advanced) • Deep Learning (Advanced) • Computer Vision (Advanced) • Statistical Analysis (Advanced) • Data Analysis (Intermediate) • Predictive Modeling (Intermediate) • Model Evaluation (Intermediate)

Research & Analytical Tools: LaTeX • Research Methodology • Experimental Design • Scientific Writing • Performance Benchmarking

Business & Productivity Tools: Microsoft Excel • Microsoft Word • Microsoft PowerPoint • GitHub

Professional Competencies: Project Management • Leadership • Team Collaboration • Problem Solving • Technical Communication • Presentation Skills

Soft Skills: Strong Communication Skills • Interpersonal Skills • Team Collaboration • Leadership • Time Management • Attention to Detail • Self-Motivated • Career-Oriented • Customer-Focused Mindset • Adaptability

CERTIFICATIONS & TRAINING

- **Advanced Computer Application Course** — National Youth & Technical Training Center (2017)
- **Data Analysis with Python** — EDGE Project, Bangladesh Computer Council, ICT Division (2025)
- **CS403: Introduction to Modern Database Systems** — Saylor Academy (2025)
- **Simplilearn SkillUp (2025):** Generative AI for Beginners • Introduction to Basic Finance • Project Management 101 • Introduction to MS Excel

EDUCATION

- **Bachelor of Science (Honours) in Mathematics** 2022 – 2026
- *Jashore University of Science and Technology (JUST)*
- **Thesis:** *Benchmarking Deep Learning Models for Robust Deepfake Video Detection*
- Supervisor: Dr. Md. Farhad Bulbul • CGPA: 2.90 / 4.00

- **Higher Secondary Certificate (Science)** 2020
- Sabujbagh Government College, Dhaka • GPA: 5.00 / 5.00

RESEARCH EXPERIENCE

Research Project — Benchmarking Deep Learning Models for Robust Deepfake Video Detection 1 Year

Jashore University of Science and Technology (JUST)

- Developed a deepfake video detection framework capable of reliably distinguishing authentic from manipulated video content.
- Benchmarked multiple CNN-based and Vision Transformer (ViT) architectures through rigorous, reproducible comparative analysis.
- Owned the full experimental pipeline — data preprocessing, model training, hyperparameter optimization, and evaluation — ensuring reproducibility and result validity.
- Achieved a **99.02% AUC** with a ViT-B/16 architecture, establishing the superiority of transformer-based methods for this task.
- Compiled and structured findings for peer-reviewed academic publication.

SELECTED RESEARCH ACHIEVEMENTS

- Authored and co-authored three research manuscripts spanning deep learning, computer vision, knowledge distillation, and scientific machine learning, currently under review at IEEE and Wiley venues.

- Presented three research posters at university and national academic conferences, communicating complex technical findings to interdisciplinary audiences.
- Applied diverse advanced methodologies — Vision Transformers, CNNs, Physics-Informed Neural Networks, and Knowledge Distillation — to build reproducible research pipelines and translate experimental results into publication-ready findings.

PUBLICATIONS

Progressive Dual-Teacher Knowledge Distillation with Domain-Aware Training for Efficient and Generalizable DeepFake Detection
IEEE (Journal) • Status: Under Review • Role: First Author

Deepfake Detection: A Comparative Analysis of CNN and Transformer-Based Methods
Wiley (Journal) • Status: Under Review • Role: First Author

Regime-Dependent Performance Analysis of Fractional Physics-Informed Neural Networks for Reaction–Diffusion Drug Delivery Systems
International Conference (UK) • Status: Under Review • Role: Co-Author

INDUSTRIAL VISIT

Square Pharmaceuticals Ltd: Observed large-scale pharmaceutical manufacturing, quality assurance practices, and operational workflows, enhancing understanding of industry standards and production management.

PROFESSIONAL EXPERIENCE

Private Tutor ≈ 4 Years

Provided personalized mathematics and science tutoring, developing tailored learning strategies that enhanced students' academic performance, problem-solving abilities, and confidence while demonstrating strong communication and mentoring skills.

LEADERSHIP EXPERIENCE

Treasurer — Dhaka Association
Jashore University of Science and Technology (JUST)

Oversaw budgeting, financial record-keeping, and event resource planning while fostering effective collaboration between student stakeholders and university administration.

LANGUAGES

Bangla — Native **English** — Professional Working Proficiency **Hindi** — Conversational Proficiency

POSTER PRESENTATIONS

Image-Based Toxicity Classification Using Vision Transformers on 2D Molecular Structure Diagrams
Mathematics in Modern World: Interdisciplinary Applications, Dept. of Mathematics, JUST — 25 June 2025

Bridging Cheminformatics and Computer Vision: Toxicity Prediction of 2D Molecular Structures Using Vision Transformers
National Pharma Conference on Bridging Research and Practice, Dept. of Pharmacy, JUST — 19 July 2025

Global-Context Vision Transformers for Attention-Based Deepfake Detection from Video Frames
University Day 2026, Research Cell, JUST — 25 January 2026

REFERENCES

Dr. Md. Farhad Bulbul
Associate Professor, Department of Mathematics, Jashore University of Science and Technology
Email: **farhad.math04@gmail.com** • Relationship: Thesis Supervisor